

CUSTOMER PURCHASE PREDICTION USING LOGISTIC REGRESSION AND RANDOM FOREST

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**MACHINE LEARNING
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Abstract

Customer Purchase Prediction Using Logistic Regression and Random Forest is a machine-learning project developed using Python to predict whether a customer is likely to make a purchase. The system uses demographic and behavioural features including age, gender, annual income, previous purchases, website visits, pages viewed, time spent on the website, emails opened, cart additions and days since the last purchase. The dataset initially contains 500 customer records and 13 columns. Data preprocessing includes duplicate removal, identifier removal, missing-value treatment, categorical encoding and numerical scaling. Logistic Regression and Random Forest are trained using leakage-safe pipelines. A tuned Random Forest is also evaluated using 5-fold GridSearchCV. The models are compared using accuracy, precision, recall, F1-score and ROC-AUC. On the held-out test set, Logistic Regression achieved 69.39% accuracy and 40.00% F1-score, while Random Forest achieved 68.37% accuracy and 31.11% F1-score. The project demonstrates a complete supervised machine-learning workflow for customer purchase prediction.

Keywords: customer purchase prediction; Logistic Regression; Random Forest; classification; customer behaviour; machine learning.

1. Introduction

1.1 Purpose of the Project

Customer purchase prediction is a machine-learning application used to estimate whether a customer is likely to purchase a product or service. Such predictions can support customer targeting, campaign planning and prioritization of potential buyers.

1.2 Problem Statement

The problem is a binary classification task. Given customer demographic and behavioural information, the system predicts the target variable Purchase, where 0 represents no purchase and 1 represents a purchase.

1.3 Existing Methods and Disadvantages

Traditional customer targeting may rely on manual rules or basic segmentation. These approaches may not capture interactions among several customer characteristics and behavioural variables. A machine-learning classifier can learn patterns from historical customer records and provide a consistent prediction.

1.4 Proposed System

The proposed system loads the customer dataset, checks data quality, removes duplicate records and the Customer_ID identifier, handles missing values, performs exploratory data analysis, creates leakage-safe preprocessing pipelines, trains Logistic Regression and Random Forest models, tunes Random Forest hyperparameters and demonstrates a final prediction for a new customer.

1.5 Research Objectives

- Prepare and describe the customer purchase dataset.
- Perform systematic preprocessing and exploratory data analysis.
- Remove identifier information and prevent data leakage.
- Develop and train Logistic Regression and Random Forest classifiers.
- Perform hyperparameter tuning and compare model performance.
- Demonstrate purchase prediction on a new customer record.

2. Literature Survey

The literature review was expanded to ten studies directly related to customer purchase prediction, online purchase behaviour, e-commerce clickstream analysis and machine-learning classification. The methodology and findings are summarized below. The table follows the same structure as the reference report.

S.No.	Title of the paper	Authors / Journal & Year	Methodology Adapted	Key Findings	Gaps
1	Modeling Purchase Behavior at an E-Commerce Web Site: A Task-Completion Approach	Catarina Sismeiro, Randolph E. Bucklin — Journal of Marketing Research, 2004	Sequential task-completion modelling using clickstream data and binary probit models.	Browsing and navigational behaviour can predict task completion and help identify likely buyers early.	Uses a specific automotive website and sequential task framework; not directly a general tabular customer classifier.
2	Purchase Prediction Using Tmall-Specific Features	Y. Zhao, L. Yao, Y. Zhang — Concurrency and Computation: Practice and Experience, 2016	Extracted click, purchase, collect and cart feature groups from Tmall activity; threshold-moving for prediction.	Combining the three feature groups provided useful purchase-prediction information; collect-and-cart features were particularly important.	Tmall-specific features and user-brand prediction limit direct transfer to small customer-level datasets.
3	A Machine Learning Framework for Customer Purchase Prediction in the Non-Contractual Setting	A. Martínez, C. Schmuck, S. Pereverzyev Jr., C. Pirker, M. Haltmeier — European Journal of Operational Research, 2020	Dynamic customer features from previous purchases; Logistic Lasso, extreme learning machine and gradient tree boosting.	Gradient tree boosting achieved 89% accuracy and 0.95 AUC for next-month purchase prediction on the studied dataset.	Requires longitudinal transaction history and a defined future purchase window.
4	Towards Early Purchase Intention Prediction in Online Session Based Retailing Systems	Ramazan Esmeli, Mohamed Bader-El-Den, Hassana Abdullahi — Electronic Markets, 2021	Session-based online behavioural features for early purchase-intention prediction.	Shows the value of predicting purchase intention before the end of a session for e-commerce decision support.	Session-level setting differs from the demographic and customer-history structure of this project.
5	On the Platform but Will They Buy? Predicting Customers' Purchase Behavior Using Deep Learning	Neha Chaudhuri, et al. — Decision Support Systems, 2021	Deep neural networks using platform engagement and customer characteristics; compared with Decision Tree, Random Forest, SVM and ANN.	Deep learning outperformed the compared conventional ML methods on the studied e-commerce dataset and highlighted platform/customer predictors.	More complex deep-learning architecture and large session dataset are beyond the scope of this compact project.
6	Customer Purchase Prediction from the Perspective of Imbalanced Data: A Machine Learning Framework Based on Factorization Machine	Research article — Expert Systems with Applications, 2021	Improved factorization-machine feature combinations with Random Forest and cost-sensitive learning for imbalanced purchases.	Addresses sparse behavioural features and class imbalance through feature combination and cost-sensitive learning.	The proposed framework is more complex than the two-model baseline used in this project.
7	Online Purchase Prediction via Multi-Scale Modeling of Behavior Dynamics	Chao Huang, Xian Wu, Xuchao Zhang, Chuxu Zhang, Jiashu Zhao, Dawei Yin, Nitesh Chawla — ACM SIGKDD, 2019	Graph Multi-Scale Pyramid Networks with convolution-recurrent modelling of temporal and category dependencies.	Models complex temporal behaviour and category dependencies for online purchase forecasting.	Requires sequential interaction data and deep neural architectures rather than compact customer-level tabular data.
8	Research on E-Commerce Purchase Prediction Model Based on Machine Learning	Zaixin Lin, Zihui Huang, Huiyan He, Jiazi Liang, Xinbei Zheng, Jingde Huang — 2023	Random Forest decision model using features from customer historical purchase data.	Random Forest was reported as a practical model for predicting customer purchasing behaviour.	The study focuses on Random Forest and does not provide the same Logistic Regression versus Random Forest comparison used here.
9	E-Commerce User Purchase Prediction Based on Improved Machine Learning Algorithms	IEEE SMART GENCON, 2023	Markov-chain-based purchase-behaviour evaluation combined with machine-learning analysis; addresses incomplete and imbalanced data.	Improved machine-learning analysis was reported to outperform the referenced k-NN approach under the study's evaluation criteria.	Uses a different modelling framework and simulation setting from this project's customer-level dataset.
10	Predicting Online Customer Purchase: The Integration of Customer Characteristics and Browsing Patterns	Decision Support Systems, 2024	Combines RFM customer characteristics with browsing-pattern and clickstream information for session-level prediction and segmentation.	Integrating customer characteristics and browsing patterns can improve purchase prediction and support personalized marketing.	Requires richer clickstream and transaction histories than the available 500-record dataset.

2.1 Literature Review Summary

The reviewed studies show three recurring directions: (1) customer and browsing features are useful for purchase prediction, (2) tree-based and ensemble methods can model nonlinear customer behaviour, and (3) richer sequential or clickstream data can improve prediction but also increases modelling complexity. The present project focuses on a compact, interpretable supervised-learning workflow using Logistic Regression and Random Forest with demographic and behavioural customer features.

3. Methodology

3.1 Proposed System Flow Diagram



Figure 1. Proposed System Flow Diagram.

3.2 Dataset Description

The dataset contains 500 customer records and 13 columns. It includes customer demographics, income, purchase history, website engagement and the binary Purchase target. Customer_ID is an identifier and is removed before model training.

Property	Value
Records	500
Total columns	13
Target	Purchase
Original class distribution	334 no purchase; 166 purchase
Duplicate rows	13
After duplicate removal	487 rows
Missing values after duplicate removal	28 cells
Modelling predictors	11 after removing Customer_ID

3.3 Experimental Setup

Component	Configuration
Environment	Jupyter Notebook
Operating system	Windows
Python	Python 3.x
Libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
Train/Test split	80% training, 20% testing; stratified
Random state	42

3.4 Data Preprocessing

- Data inspection, schema validation, missing-value checking and duplicate checking were performed.
- 13 duplicate rows were removed, leaving 487 records.
- Customer_ID was removed because it is an identifier rather than a behavioural predictor.
- Numerical missing values were handled using median imputation.
- Categorical missing values in Gender and Customer_Segment were handled using most-frequent imputation.
- Categorical variables were one-hot encoded.
- Numerical variables were standardized for Logistic Regression.
- Preprocessing was implemented inside the model pipeline to prevent information leakage from the test set.

3.5 Exploratory Data Analysis

EDA included descriptive statistics, purchase-class distribution, age and annual-income distributions, categorical distributions, boxplots and a correlation heatmap. After duplicate removal, approximately 66.5% of records represented no purchase and 33.5% represented a purchase. No single numerical feature showed strong linear correlation with Purchase.

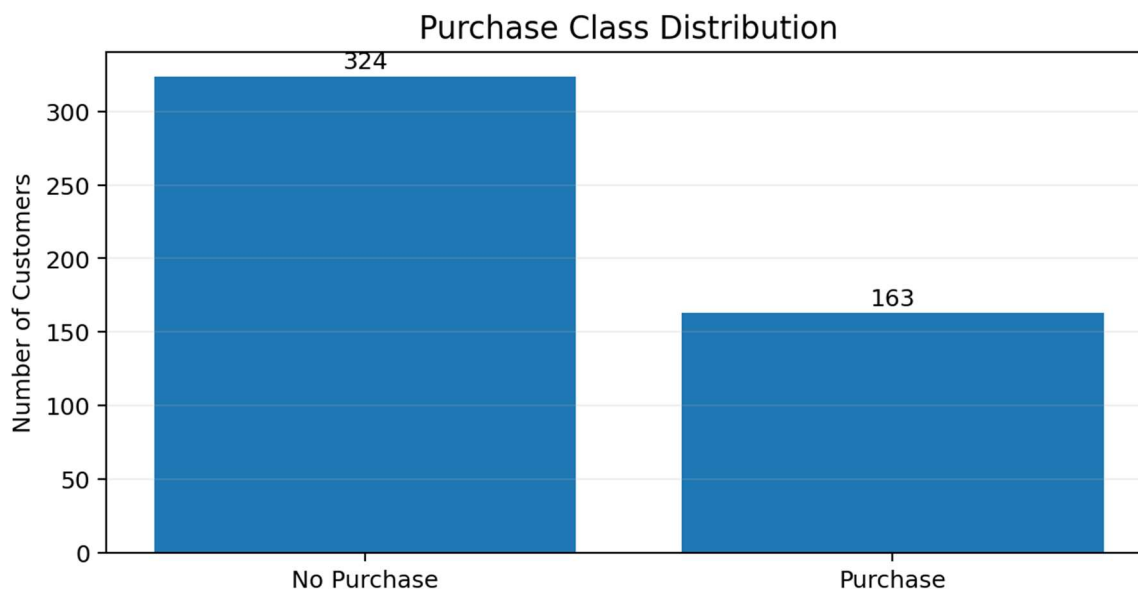


Figure 2. Purchase-class distribution after duplicate removal.

3.6 Feature Engineering and Feature Selection

No artificial features were required for the final model. Customer_ID was excluded because it is an identifier. The final predictor set consisted of Age, Gender, Annual_Income, Customer_Segment, Previous_Purchases, Website_Visits, Pages_Viewed, Time_on_Website_Min, Emails_Opened, Cart_Additions and Days_Since_Last_Purchase. Purchase was used as the target.

3.7 Model Building and Training

Model	Purpose	Configuration
Logistic Regression	Linear baseline for binary classification	max_iter=1000; preprocessing pipeline; standardized numerical variables; one-hot categorical variables
Random Forest	Nonlinear ensemble classifier	n_estimators=100; random_state=42; preprocessing pipeline

3.8 Hyperparameter Tuning

GridSearchCV with 5-fold cross-validation was applied to the Random Forest pipeline. The search considered n_estimators, max_depth and min_samples_split. Accuracy was used as the cross-validation scoring metric.

Tuning Item	Value
Search method	GridSearchCV
Cross-validation	5-fold
Scoring metric	Accuracy
Search parameters	n_estimators, max_depth, min_samples_split
Best parameters	n_estimators=50, max_depth=10, min_samples_split=5
Best CV accuracy	69.17%

3.9 Model Evaluation

The test set was held out from model fitting. Accuracy, precision, recall, F1-score, ROC-AUC, classification reports and confusion matrices were used. ROC-AUC was calculated using predicted probabilities.

4. Results and Discussion

4.1 Preprocessing Results

Stage	Result
Initial dataset	500 rows and 13 columns
Duplicate check	13 duplicate rows removed
After duplicate removal	487 rows
Identifier removal	Customer ID removed
Missing-value treatment	Median imputation for numerical; most-frequent for categorical
Final modelling data	487 rows with 11 predictors and 1 target

4.2 Model Performance

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	69.39%	58.82%	30.30%	40.00%	0.6755
Random Forest	68.37%	58.33%	21.21%	31.11%	0.6823
Tuned Random Forest	65.31%	46.67%	21.21%	29.17%	0.6135

Logistic Regression achieved the highest held-out test accuracy and F1-score among the three models. Random Forest achieved the highest ROC-AUC. The tuned Random Forest improved its cross-validation accuracy during model selection, but its held-out test performance was lower than the original Random Forest on this split.

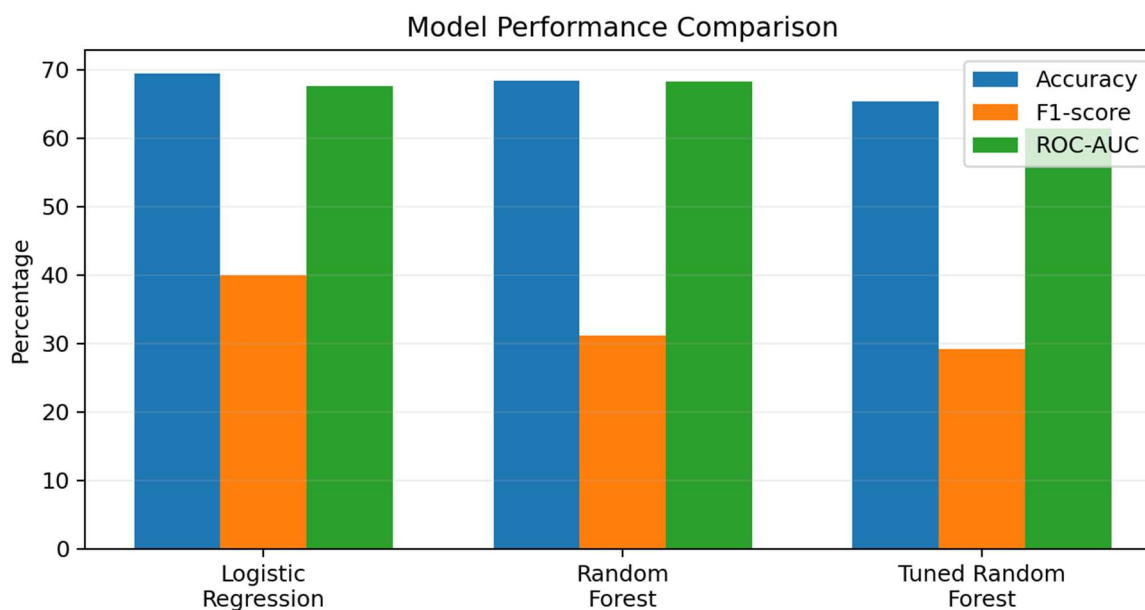


Figure 3. Model performance comparison.

4.3 Confusion Matrix Results

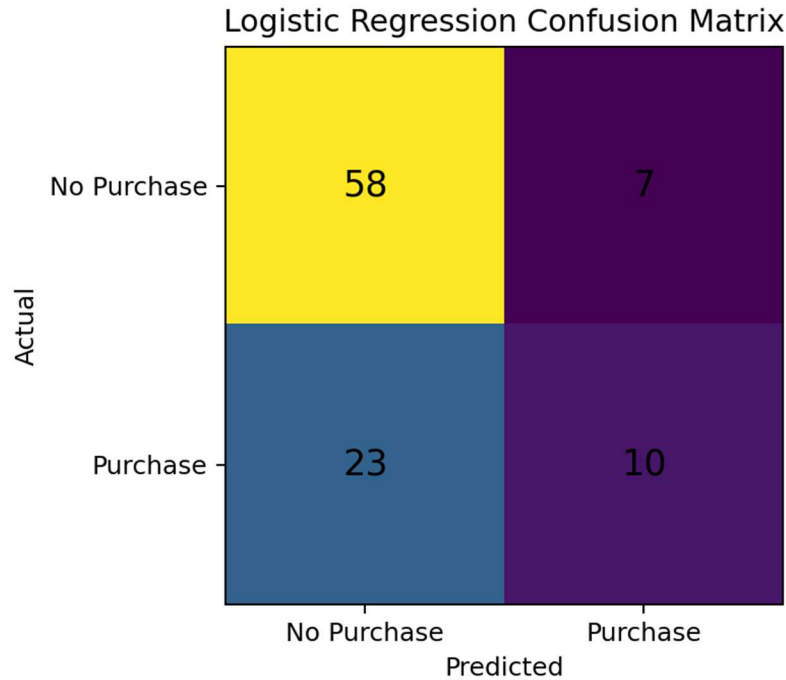


Figure 4. Confusion matrix for Logistic Regression.

The Logistic Regression confusion matrix was $[[58, 7], [23, 10]]$, corresponding to 58 true negatives, 7 false positives, 23 false negatives and 10 true positives. The model identified no-purchase customers more effectively than purchase customers.

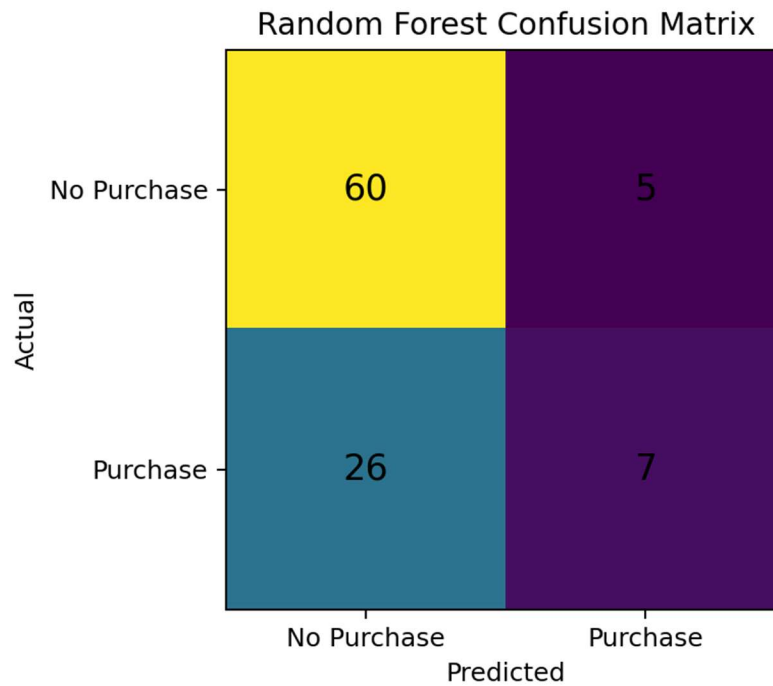


Figure 5. Confusion matrix for Random Forest.

The Random Forest confusion matrix was $[[60, 5], [26, 7]]$, corresponding to 60 true negatives, 5 false positives, 26 false negatives and 7 true positives. The model detected fewer actual purchasers than no-purchase customers.

4.4 Random Forest Feature Importance

Feature	Importance
Days_Since_Last_Purchase	0.1677
Annual_Income	0.1409
Time_on_Website_Min	0.1090
Website_Visits	0.1028
Age	0.1024
Emails_Opened	0.0861
Pages_Viewed	0.0821
Previous_Purchases	0.0697
Cart_Additions	0.0657

The fitted Random Forest assigned relatively higher importance to Days_Since_Last_Purchase, Annual_Income, Time_on_Website_Min, Website_Visits and Age. Feature importance indicates contribution to model predictions and does not establish a causal relationship.

4.5 Results and Discussion

Three classification models were implemented and evaluated: Logistic Regression, Random Forest and Tuned Random Forest. Logistic Regression achieved an accuracy of 69.39%, precision of 58.82%, recall of 30.30%, F1-score of 40.00% and ROC-AUC of 0.6755. Random Forest achieved 68.37% accuracy, 58.33% precision, 21.21% recall, 31.11% F1-score and 0.6823 ROC-AUC. The tuned Random Forest achieved 69.17% cross-validation accuracy but 65.31% test accuracy, with 46.67% precision, 21.21% recall, 29.17% F1-score and 0.6135 ROC-AUC. The models classified no-purchase customers more effectively than purchase customers, as shown by the lower recall for class 1. The results demonstrate the differences between a linear baseline, an ensemble model and a tuned ensemble model on the given customer dataset.

5. Final Prediction / Demonstration

Input Feature	Value
Age	35
Gender	Male
Annual Income	80000
Customer Segment	Regular
Previous Purchases	5
Website Visits	10
Pages Viewed	6
Time on Website (min)	12
Emails Opened	4
Cart Additions	2
Days Since Last Purchase	60

Predicted Purchase: 0 (No Purchase) , Purchase Probability: 0.4183 (41.83%)

The trained Logistic Regression model was used to demonstrate prediction on a new customer record. At the default classification threshold, the model predicts no purchase with a purchase probability of 41.83%.

6. Conclusion

This project developed a customer purchase prediction system using Logistic Regression and Random Forest. The workflow included data inspection, duplicate removal, missing-value treatment, identifier removal, categorical encoding, numerical scaling, leakage-safe preprocessing, model training, hyperparameter tuning and multi-metric evaluation. Logistic Regression achieved 69.39% test accuracy and a 40.00% F1-score, while Random Forest achieved 68.37% accuracy and a 0.6823 ROC-AUC. The final demonstration showed how the trained Logistic Regression model can predict the purchase outcome and probability for a new customer. The project demonstrates the application of supervised machine learning to customer behaviour analysis and purchase prediction.

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Signature of the Guide: _____ **Course Instructor:** _____